

# Alejandro Sánchez Yalí

## AI Full Stack Engineer



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## SUMMARY

AI Full Stack Engineer with 8+ years designing and shipping web applications end to end and 3+ years building LLM-powered products. Architect and lead engineer of three deployed systems – an AI WhatsApp support platform, an AI-assisted recruitment platform and a surgical supply chain system – spanning Python/FastAPI backends, Next.js/TypeScript frontends, PostgreSQL and cloud deployment. M.Sc. in Mathematics and 11 years teaching mathematics and physics, which shows up in design docs and code review.

## TECHNICAL SKILLS

- **Languages:** Python (8+ years), TypeScript (6+ years), JavaScript (6+ years), SQL, Rust (personal projects)
- **Frontend:** React.js (7+ years), Next.js (6+ years), Tailwind CSS, Shadcn UI, Redux, Zustand, Zod, Effect, Three.js
- **Backend:** FastAPI, Django, Django REST Framework, Celery, Flask, NestJS, Node.js, Express, RESTful APIs, GraphQL, microservices
- **AI & LLMs:** LangChain, LangGraph, LiteLLM, RAG, MCP, OpenAI API, Whisper, ElevenLabs, PyTorch, TensorFlow
- **Databases & Infra:** PostgreSQL, Redis, MongoDB, MySQL, Firebase, Docker, AWS, GCP, GitHub Actions, CI/CD
- **Engineering:** Clean architecture, DDD, SOLID, TDD, Vitest, Jest, React Testing Library, code review

## EXPERIENCE

### AI Specialist

*Lapzo*

Aug 2025 – Sep 2026

*Remote, Mexico*

- Designed and built a Digital Professor that holds real-time conversations with learners and explains course content through interactive slides
- Owned the architecture connecting LLM services, ElevenLabs voice synthesis and the real-time layer built with Next.js, NestJS and Firebase Realtime Database
- Closed 57 tasks and 6 epics in the first nine months of 2026 with a single production bug, per the team's board
- Built the AI content-generation pipelines (LangChain, LangGraph, n8n), with Celery workers running generation in the background

### Independent Software & AI Consultant

*Freelance*

Aug 2024 – Present

*Remote*

- Delivered end-to-end web applications for business clients across both halves of the stack: React.js/Next.js/TypeScript on the front, Python/FastAPI/Django and Node.js/NestJS behind it
- Lead engineer of Aluna (AI recruitment platform, see below) and builder of a multi-tenant AI customer service agent for CREARIA (Colombia) on WhatsApp Cloud API, LiteLLM, RAG and MCP with human escalation
- Ran technical design sessions with clients and made the technology decisions for each project; code reviewer for a client's development team
- Deployed on AWS (EC2, S3, Docker) with CI/CD pipelines and Stripe payment integrations

### Full Stack Engineer

*Monadical*

Apr 2021 – Aug 2024

*Remote, Canada/US*

- Full-stack engineer on ZoHuddle, a video conferencing app for scientific conferences (Neuromatch) that matches participants by mutual interests: React front end and Django back end, with Django Channels (WebSockets) and Twilio Video
- Full-stack engineer on OddSlingers, an in-browser poker platform for desktop and mobile: React front end and Django back end on an event-stream architecture
- Developed both frontend and backend, including reusable components and tests (Vitest, Jest, TDD)
- Integrated AI/ML services and LLMs (OpenAI, LangChain, Whisper) into production web apps
- Did code reviews on a distributed team across Canada, the US and Latin America

## Full Stack Engineer

BCFort

Aug 2018 – Oct 2020

Medellín, Colombia

- Designed system architecture for blockchain platforms and built decentralised app frontends with React and Web3.js
- Developed analytics platforms with Python (Django) and Node.js, managing cloud infrastructure on AWS, GCP and IBM Cloud

## Professor of Mathematics & Physics (part-time)

Universidad de Antioquia

Jan 2010 – Oct 2021

Medellín, Colombia

- Taught mathematics and physics; in the last year and a half also machine learning and data science with Python, NumPy, Pandas, TensorFlow and PyTorch
- Developed the AI/ML elective curriculum for the Mathematics Institute and mentored student research projects

## FEATURED PROJECTS

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- **Plixiq – AI Customer Support on WhatsApp**  – Python, FastAPI, LiteLLM, PostgreSQL, Redis, Next.js, WebSocket
  - Architect and lead engineer of a multi-tenant SaaS where AI agents answer routine customer inquiries, book appointments and escalate the rest to human operators with full conversation context
  - Modular monolith (14 components plus a shared kernel) with boundaries enforced in CI; LiteLLM layer with a primary and fallback provider per agent, so an outage becomes a fallback
- **VitaStock – Surgical Supply Chain for Clinics**  – Python, FastAPI, Next.js, TypeScript, PostgreSQL, Docker, Railway
  - Architect and lead engineer of a platform commissioned by a private surgical clinic in Colombia – 12 modules and 5 user roles designed to replace spreadsheet-based inventory and procurement; delivered, with rollout at the clinic pending
  - Surgery scheduling with supply baskets and real-time consumption tracking, lot-level traceability with expiry alerts, and true per-patient cost through a cardex
- **Aluna – AI Recruitment Platform for Staffing Agencies**  – Next.js 15, FastAPI, LiteLLM, PostgreSQL, Inngest, Claude, Gemini, WhatsApp Cloud API
  - Lead engineer of an AI recruitment platform for staffing agencies, built in 3 months – 336 source files across 4 packages, 26 tables and 11 external providers – split into a Next.js 15 application owning all domain state and a stateless FastAPI service owning model calls and the WhatsApp channel
  - Durable Inngest analysis pipeline (9 checkpointed steps, 3 retries) where deduplication runs before quota enforcement so a re-analysed CV reuses its cached score instead of paying for another model call; one screening engine serves WhatsApp, web chat and email

## OPEN SOURCE

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- **talking-avatar-with-ai**  – 464 stars, 110 forks. Open-source conversational AI avatar: OpenAI GPT, Whisper, ElevenLabs, Rhubarb lip-sync, Three.js
- **Monadical-SAS/Morpheus**  – 19 pull requests. Open-source AI image generation platform: FastAPI, React, Stable Diffusion
- **kipux**  – Collaborative-finance app in Rust (Axum, sqlx, PostgreSQL, HTMX, SSE, PWA) with Claude-vision receipt parsing. Open source (MIT), work in progress

## TALKS & WRITING

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- **Scaling Learning Models: Parallelism Strategies in JAX with Flax**  – PyCon Colombia, 2024
- **Building Plixiq: a Multi-Tenant WhatsApp AI Support Platform** – architecture case study (2026)

## EDUCATION

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### M.Sc. in Mathematics

Universidad de Antioquia

Aug 2010 - Sep 2013

Medellín, Colombia

### B.Sc. in Teaching Mathematics and Physics

Universidad de Antioquia

Jan 2004 - Oct 2009

Medellín, Colombia